Electrostatics

Coulomb's law

Dr. Sampath Kumar

Associate Professor

Department of ECE

MIT, Manipal

The Experimental Law of Coulomb

- Coulomb stated that the force between two very small objects separated in vacuum or free space by a distance which is large compared to their size is proportional to the charge on each and inversely proportional to the square of the distance between them

$$F = k\frac{Q_1 Q_2}{R^2}$$

- In SI Units, the quantities of charge Q are measured in coulombs (C), the separation R in meters (m), and the force F should be newtons (N)
- This will be achieved if the constant of proportionality k is written as:

$$k = \frac{1}{4\pi\varepsilon_0}$$

The Experimental Law of Coulomb

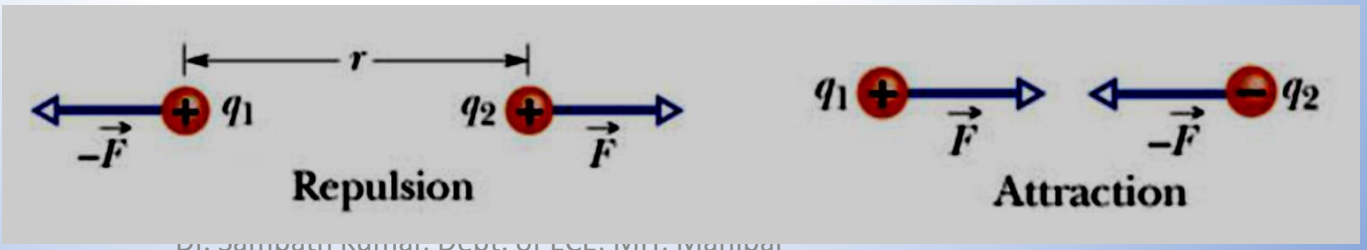
- The permittivity of free space ε_0 is measured in farads per meter (F/m), and has the magnitude of:

$$\varepsilon_0 = 8.854 \times 10^{-12} \doteq \frac{1}{36\pi} 10^{-9} \text{ F/m}$$

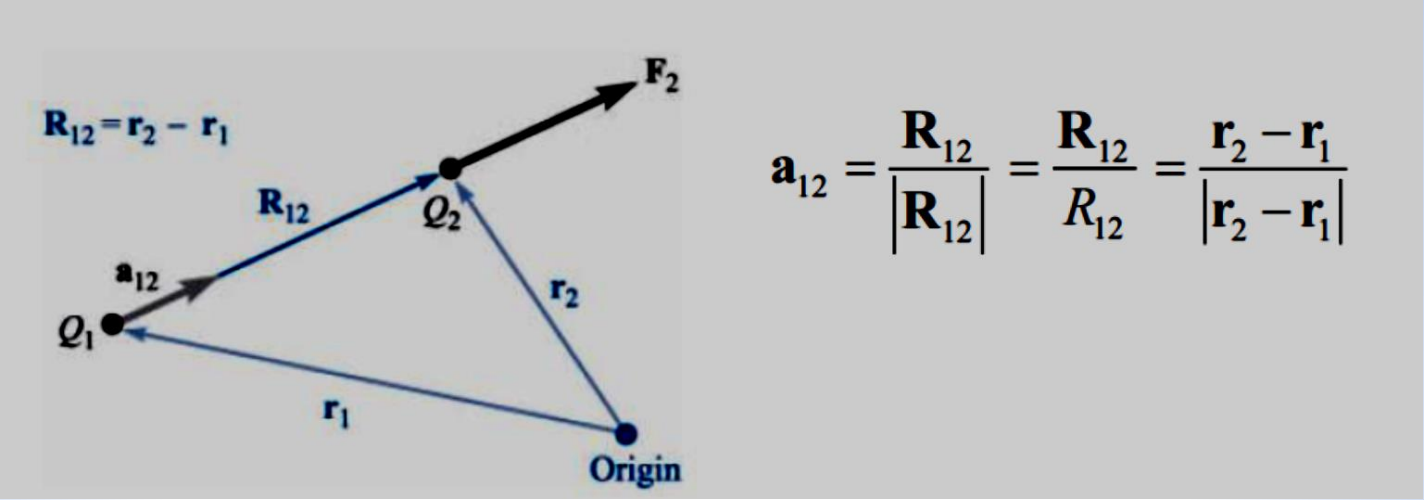
- The Coulomb's law is now:

$$F = \frac{1}{4\pi\varepsilon_0} \frac{Q_1 Q_2}{R^2}$$

- The force F acts along the line joining the two charges
- It is repulsive if the charges are alike in sign and attractive if they are of opposite sign.



The Experimental Law of Coulomb



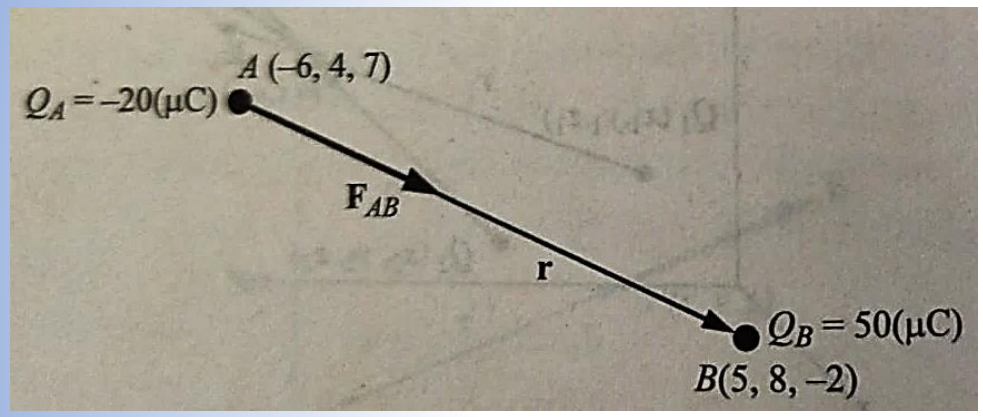
$$\mathbf{a}_{12} = \frac{\mathbf{R}_{12}}{|\mathbf{R}_{12}|} = \frac{\mathbf{R}_{12}}{R_{12}} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|}$$

- In vector form, Coulomb's law is written as:

$$\mathbf{F}_2 = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{R_{12}^2}\mathbf{a}_{12}$$

- F_2 is the force on Q_2, for the case where Q_1 and Q_2 have the same sign,
- While a_{12} is the unit vector in the direction of R_{12}, the line segment from Q_1 to Q_2.

D2.1. A charge $Q_A = -20\ \mu C$ is located at $A(-6, 4, 7)$, and a charge $Q_B = 50\ \mu C$ is at $B(5, 8, -2)$ in free space. If distances are given in meters, find: (*a*) $\mathbf{R}_{AB}$; (*b*) R_{AB}. Determine the vector force exerted on Q_A by Q_B if $\epsilon_0 =$ (*c*) $10^{-9}/(36\pi)$ F/m; (*d*) 8.854×10^{-12} F/m.



a. $\mathbf{r} = [5 - (-6)]\,\mathbf{a}_x + [8 - 4]\,\mathbf{a}_y + [-2 - 7]\,\mathbf{a}_z$

$= 11\,\mathbf{a}_x + 4\,\mathbf{a}_y - 9\,\mathbf{a}_z$

b. $r = |\mathbf{r}| = \sqrt{11^2 + 4^2 + (-9)^2} = 14.76$ (m)

$$\mathbf{a}_r = \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z}{14.76}$$

$$\mathbf{F}_{AB} = \frac{|Q_A||Q_B|}{4\pi\epsilon r^2}\,\mathbf{a}_r$$

$$\Rightarrow \quad \mathbf{F}_{AB} = \frac{20 \times 10^{-6} \times 50 \times 10^{-6}}{4\pi \times 8.854 \times 10^{-12} \times 1 \times (14.76)^2}\left[\frac{11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z}{14.76}\right]$$

Please note that, we have taken $\epsilon_r = 1$ in the evaluation of the above expression.

Therefore, $\mathbf{F}_{AB} = 30.75\,\mathbf{a}_x + 11.18\,\mathbf{a}_y - 25.16\,\mathbf{a}_z$ (mN).

Note: $\varepsilon = \varepsilon_0\varepsilon_r$

Electric Field Intensity

- Let us consider one charge, say Q_1, fixed in position in space
- Now, imagine that we can introduce a second charge, Q_t, as a "unit test charge", that we can move around
- We know that there exists everywhere a force on this second charge
- This second charge is displaying the existence of a force field
- The force on it is given by Coulomb's law as:

$$\mathbf{F}_t = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_t}{R_{1t}^2}\mathbf{a}_{1t}$$

- Writing this force as a "force per unit charge" gives:

$$\frac{\mathbf{F}_t}{Q_t} = \frac{1}{4\pi\varepsilon_0}\frac{Q_1}{R_{1t}^2}\mathbf{a}_{1t}$$

Vector Field,
Electric Field Intensity

Electric Field Intensity

- We define the electric field intensity as the vector of force on a unit positive test charge
- Electric field intensity, E, is measured by the unit newtons per coulomb (N/C) or volts per meter (V/m)

$$\mathbf{E} = \frac{\mathbf{F}_t}{Q_t} = \frac{1}{4\pi\varepsilon_0}\frac{Q_1}{R_{1t}^2}\mathbf{a}_{1t}$$

- The field of a single point charge can be written as:

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\mathbf{a}_R$$

- $\mathbf{a}_R$ is a unit vector in the direction from the point at which the point charge Q is located, to the point at which E is desired/measured.

Electric Field Intensity

- For a charge which is not at the origin of the coordinate, the electric field intensity is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{|\mathbf{r}-\mathbf{r}'|^2}\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{Q(\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|^3}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{Q\left[(x-x')\mathbf{a}_x + (y-y')\mathbf{a}_y + (z-z')\mathbf{a}_z\right]}{\left[(x-x')^2 + (y-y')^2 + (z-z')^2\right]^{3/2}}$$

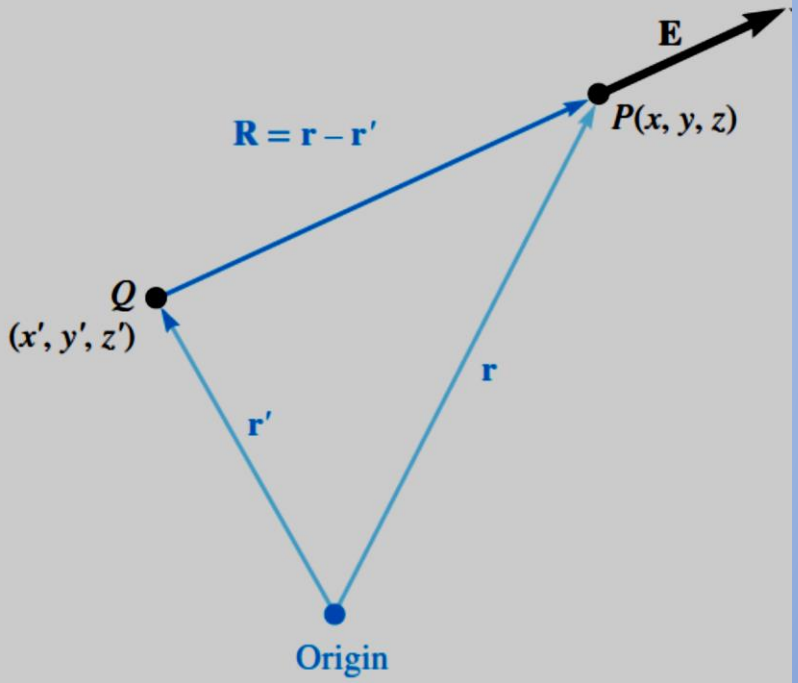
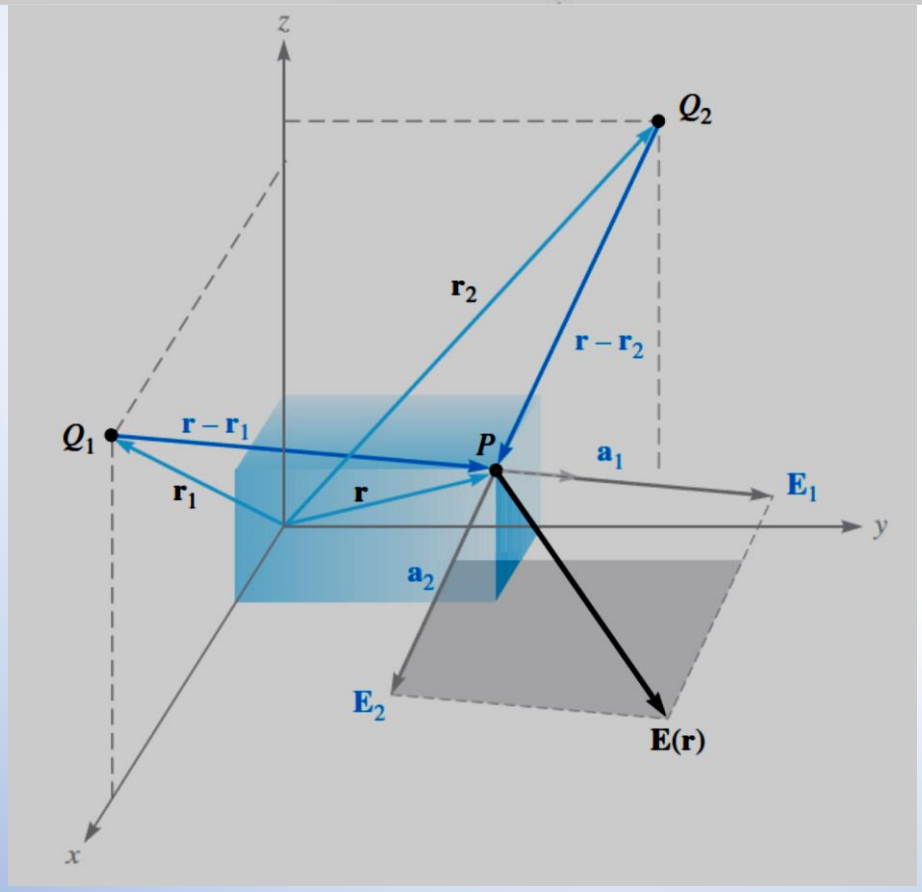


Figure 2.2 The vector **r′** locates the point charge *Q*, the vector **r** identifies the general point in space *P*(*x*, *y*, *z*), and the vector **R** from *Q* to *P*(*x*, *y*, *z*) is then **R** = **r** − **r′**.

Electric Field Intensity

- The electric field intensity due to two point charges, say Q_1 at r_1 and Q_2 at r_2, is the **sum of the electric field intensity on Q_t** caused by Q_1 and Q_2 acting alone (Superposition Principle).



$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{|\mathbf{r}-\mathbf{r}_1|^2}\mathbf{a}_1 + \frac{1}{4\pi\varepsilon_0}\frac{Q}{|\mathbf{r}-\mathbf{r}_2|^2}\mathbf{a}_2$$

Electric Field Intensity

■ Example

A charge Q_1 of 2 μC is located at at $P_1(0,0,0)$ and a second charge of 3 μC is at $P_2(-1,2,3)$. Find E at $M(3,-4,2)$.

$$\mathbf{r}-\mathbf{r}_1 = 3\mathbf{a}_x - 4\mathbf{a}_y + 2\mathbf{a}_z, \quad |\mathbf{r}-\mathbf{r}_1| = \sqrt{29}$$

$$\mathbf{r}-\mathbf{r}_2 = 4\mathbf{a}_x - 6\mathbf{a}_y - \mathbf{a}_z, \quad |\mathbf{r}-\mathbf{r}_2| = \sqrt{53}$$

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{Q_1}{|\mathbf{r}-\mathbf{r}_1|^2}\mathbf{a}_1 + \frac{1}{4\pi\varepsilon_0}\frac{Q_2}{|\mathbf{r}-\mathbf{r}_2|^2}\mathbf{a}_2$$

$$= \frac{1}{4\pi\varepsilon_0}\left\{\frac{Q_1(\mathbf{r}-\mathbf{r}_1)}{|\mathbf{r}-\mathbf{r}_1|^3} + \frac{Q_2(\mathbf{r}-\mathbf{r}_2)}{|\mathbf{r}-\mathbf{r}_2|^3}\right\}$$

$$= \frac{1}{4\pi\varepsilon_0}\left\{\frac{2\times10^{-6}(3\mathbf{a}_x - 4\mathbf{a}_y + 2\mathbf{a}_z)}{|\sqrt{29}|^3} + \frac{3\times10^{-6}(4\mathbf{a}_x - 6\mathbf{a}_y - \mathbf{a}_z)}{|\sqrt{53}|^3}\right\}$$

$$= \underline{\underline{623.7\mathbf{a}_x - 879.92\mathbf{a}_y + 160.17\mathbf{a}_z \ \text{V/m}}}$$

Try this

D2.2. A charge of $-0.3\ \mu C$ is located at $A(25, -30, 15)$ (in cm), and a second charge of $0.5\ \mu C$ is at $B(-10, 8, 12)$ cm. Find $\mathbf{E}$ at: (a) the origin; (b) $P(15, 20, 50)$ cm.

Ans. $(a)\ 92.3\mathbf{a}_x - 77.6\mathbf{a}_y - 94.2\mathbf{a}_z$ kV/m; $(b)\ 11.9\mathbf{a}_x - 0.519\mathbf{a}_y + 12.4\mathbf{a}_z$ kV/m


Thank
you